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MILITARY SPECIFICATION

DEVELOPER, PHOTOGRAPHIC, LOW CONTRAST, ARMED FORCES DEVELOPER NO. 3

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements of a processing formula for a photographic developer which is supplied in single powder dry form, properly stabilized, for use in the preparation of solutions for developing photographic films to low contrast at 68° F.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

O-C-275	Chemicals, Photographic, Single Substance, General Specification for
PPP-P-16	Packaging of Photographic Dry Chemicals, General Specification for
PPP-C-96	Can, Metal, 28 Gage and Lighter

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

Federal

FEDERAL STANDARD NO. 170	Film, Photographic, Black and White, Classification and Testing Methods
FEDERAL STANDARD NO. 595	Colors

FSC 6750

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemicals.- The chemicals used in the processing formula shall be photographic grade conforming to O-C-275. Materials used to prevent deterioration of the dry chemicals in the single-blend powder are not specifically designated but shall be entirely suitable for the purpose intended. The materials used shall be freely soluble in water, and relatively nontoxic.

3.1.1 Mixing of chemicals.- The dry chemicals in powder form shall be mixed thoroughly to form a homogeneous blend before dispensing into the unit container. The chemicals shall be free flowing, readily soluble, free from foreign matter, and shall show no signs of discoloration or dark specks. The temperature and relative humidity of the mixing room shall not exceed 75° F and 25 percent relative humidity.

3.2 Composition of formula.- The developer shall consist of low contrast film developing chemicals having the composition specified in table I. This specification covers a single-blend powder for 1- and 5- gallon sizes. Unless otherwise specified the allowable tolerance of the total net chemical weight in each unit container and the weight of the individual chemicals shall be ± 3 percent. The supplier shall certify that the chemical composition of the developer furnished conforms to the appropriate formulation specified in table I.

TABLE I. Composition of formula

Chemicals	1-gallon size (grams)	5-gallon size (grams)
Sodium sulfite	378.0	1,890.0
Sodium tetraborate	7.6	38.0
Mono-methyl-para-aminophenol sulphate	7.6	38.0
Hydroquinone	18.9	94.5
Total chemical weight 1/	1/ 412.1	1/ 2,060.5

1/ To allow this formula to be prepared in a single powder form it may be necessary to use additional water soluble components, otherwise photographically inert, to prevent interaction or decomposition of the active components while in packaged form.

3.3 Performance.- The developer shall perform equal to the control solution when tested as specified in 4.3.2.

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3.4 Unit sizes.- The chemicals shall be packaged in unit sizes sufficient to prepare working solutions of either 1 or 5 gallson as specified by the procuring activity (see 6.2). For packages other than those specified herein, the weight of the chemical composition shall be extrapolated accordingly.

3.5 Storage stability.- The dry powder in the unit package, shall show no evidence of deterioration after storage for a period of 168 hours at 130° ± 2° F.

3.6 Workmanship.- Workmanship shall be in accordance with high grade commercial practice covering the manufacture and packaging of photographic materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.- Unless otherwise specified, sampling shall be performed in accordance with the provisions set forth in MIL-STD-105. For purposes of sampling, lots shall consist of all material of the same type and item submitted for acceptance and delivery at one time.

4.3 Testing of the end item.- The photographic processing formula shall be tested for applicable characteristics as indicated in table II for each lot presented for acceptance. The lot size shall be expressed in units of primary containers of developer. The sample unit for the dissolving and performance tests shall be one primary container. The sample unit for the storage stability test shall be one primary container. The inspection levels and acceptable quality levels shall be as indicated in table II.

4.3.1 Dissolving test.- The powder obtained from each sample shall be added to 75 percent of the final volume of distilled water at 90° F while stirring with sufficient agitation to keep the chemicals suspended. Stirring shall continue until chemicals are dissolved. Sufficient water shall be added to make final volume of developer and stirring shall continue until a uniform solution is obtained. The mixing process should not exceed 10 minutes. Ten minutes after mixing, the chemical solution shall be clear. When cooled to 68° F, the developer solution shall be colorless to light pink in color. Although a slight flocculence of the cooled liquid is permissible, foreign matter in any form, rust, dirt, or undissolved chemicals shall be cause for rejection.

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TABLE II. Instructions for testing - sample units

Characteristic	Spec. reference		Rqmts. applied to		No. Determinations per sample unit	Results reported as		Inspection level 3/	AQL in defects per 100 units
	Requirement	Test method	Indiv. unit	Lot avg.		Pass or fail 1/	Numerically to nearest 2/		
Dissolving test	3.1	4.3.1	X	---	1	X	---	S-1	2.5
Performance test	3.3	4.3.2							
Difference in gamma			X	---	Avg. of 2	---	0.01	S-1	2.5
Difference in base and fog density			X	---	Avg. of 2	---	0.01	S-1	2.5
Difference in D max			X	---	Avg. of 2	---	0.01	S-1	2.5
Storage stability test	3.5	4.3.3							
Condition of powder			X	---	1	X	---	S-1	2.5
Rust			X	---	1	X	---	S-1	2.5
Performance	3.3	4.3.2							
Difference in gamma			X	---	Avg. of 2	---	0.01	S-1	2.5
Difference in base and fog density			X	---	Avg. of 2	---	0.01	S-1	2.5
Difference in D max			X	---	Avg. of 2	---	0.01	S-1	2.5

1/ If failure is indicated, report either description of failure or numerical point of failure, as applicable.
2/ Test reports shall include all values on which results are based.
3/ Not less than 7 sample units shall be tested.

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4.3.2 Performance test.- Prepare two developer solutions in distilled water. The chemicals to be used in the first solution shall be obtained from the chemicals being used in production of the developer. This mixture shall be identified as "sample solution". The chemicals used for the other solution shall be obtained from an approved ASA standard product and prepared in accordance with the formula specified in table I. The latter mixture shall be labeled "control solution". Expose identical sensitometric images, containing at least eight steps, onto each of two strips of photographic film taken from the same box or roll. Process one strip in each of the respective prepared solutions in accordance with the procedure described in Federal Standard No. 170. Compare the sensitometric characterisitics of the film developed in the solution prepared according to the specified formula. The gamma of the curves prepared from film developed in the test solution and the control solution shall not vary by more than ±0.10, base and fog density shall not vary by more than ±0.02, and D max shall not vary by more than ±0.10. Tests shall be performed in duplicate on each sample unit. The average of the two determinations on each sample unit shall be used for plotting the sensitometric curve. Appropriate procedures for exposure of the film and preparation of the sensitometric curve, as contained in Fed. Std. No. 170 shall be used. Speed method selected for each identical process sensitometric image shall not vary more than one-eighth of a stop by comparison.

4.3.3 Storage stability test.- Unit containers selected at random from each lot shall be stored at 130° F for a period of 168 hours. The samples shall then be removed and allowed to cool to room temperature for 2 hours and then examined. The powder shall be dry, white to light buff in color, free-flowing, and free from lumps. No rust shall be observed in the interior of the container. The contents of the container shall then be subjected to the performance test specified in table III and 4.3.2.

4.4 Inspection of preparation for delivery.- Sample items and packs shall be selected and inspected in accordance with PPP-P-16.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- The preparation of the developer for delivery, in 1- and 5-gallon sizes, or other size required in 3.4, shall be in accordance with PPP-P-16, except as specified herein.

5.2 Unit packaging.- Unless otherwise specified by the procuring activity, unit packaging shall be in accordance with PPP-P-16, level A. The unit package shall consist of a lithographed metal container, hermetically sealed, conforming to PPP-P-16, type II, class 1. The exterior surface of the can shall be finished in accordance with plan B of PPP-C-96, and the color shall be No. 24087, olive drab, Fed. Std. No. 595.

5.3 Unit container marking.- Unless otherwise specified by the procuring activity, the unit container shall be durably and legibly marked by a lithographic or similar process in semigloss yellow lettering, color No. 23538 of Fed. Std. No. 595, directly on the body of the container, provided the order is for at least 5,000 of the 1- or 5-gallon sizes. A paper label may be used for procurement of less than 5,000 units. Each label shall be completely coated on the back side with a waterproof latex glue or a heat-sealing resin adhesive. This waterproof adhesive shall be used to securely affix the paper labels to the exterior side of the container. The labels shall have waterproof black printing on a high grade paper stock.

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5.3.1 Each unit container shall be marked or labeled in accordance with figure 1. The mixing and size information given is for the 1- gallon size. Appropriate changes shall be made for the 5- gallon size, or other size required in 3.4.

DIRECTIONS FOR MIXING (1-gallon size)	DEVELOPER, PHOTOGRAPHIC, LOW CONTRAST ARMED FORCES DEVELOPER NO. 3
Pour the contents of this can into 3 quarts of warmed distilled water (90° F) (32° C) while stirring with sufficient agitation to keep the chemicals suspended. Add distilled water to make 1 gallon. Continue stirring until all chemicals are completely dissolved and a uniform solution is obtained. DIRECTIONS FOR USE Aerial photography - Use full strength. For low contrast, using fresh developer in equipment such as the type B-5 film developer assembly, develop high speed aerial panchromatic film, class 11-14-64k, for 16 minutes at 68° F. The time for aerial roll film is based on 75-foot lengths. When developing longer lengths of film, increase the developing times by 10 percent for each additional 25 feet of film; decrease the time by 10 percent for each 25 feet less. For more contrast, slightly increase developing time. Nonaerial photography - Use full strength. Use this developer when low contrast is desired. Average development time at 68° F is 17 minutes for tank development and 15 minutes for tray development.	FINAL VOLUME 1-gallon size (0.833 imperial gallon) (3.8 liters) Specification No. _____ Contract No. _____ Stock No. _____ Date of mfr._____ Name of mfr'r. or trademark ___ Net weight _____ PHOTOGRAPHIC CHEMICALS STORE IN A DRY COOL PLACE

FIGURE 1. Unit container directions and marking.

5.3.1.1 Date of manufacture.- The date of manufacture shall be the month and year of manufacture of the material specified herein. At the discretion of the supplier, the date of manufacture will either be lithographed directly

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on the label of the container or clearly and permanently stamped on the lid. Example: (Feb 66) would refer to February 1966.

5.4 Packing.- Packing shall be level A, B, or C of PPP-P-16 as specified in the contract or order (see 6.2).

5.5 Marking.- Marking shall be in accordance with PPP-P-16. In addition, the following precautionary marking shall be included on the exterior container:

PHOTOGRAPHIC CHEMICALS
STORE IN A COOL DRY PLACE

6. NOTES

6.1 Intended use.- The developer covered by this specification is intended for use in processing film where low contrast is required.

6.2 Ordering data.- Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Nominal size (1- or 5-gallon) unit package required (see 3.4).
(c) Quantity required.
(d) Applicable level of packing required (see 5.4).

6.3 International standardization agreement.- Certain provisions of this specification (5.3.1) are the subject of international standardization agreement (ASCC Air Std 52/3, STANAG 3187). When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:
Army - EL
Navy - AS
Air Force - 11

Reviewer activities:
Army - EL
Navy - AS, MC
Air Force - 11, 70

User activities:
Army - AV, MI
Navy - SH, YD

International interest (see 6.3)

Preparing activity:
Air Force - 11

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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